CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
4th Semester of B. Pharm. Examination
University Theory Examination May 2018
PH212 Physical Pharmaceutics

Date: 12/05/2018, Saturday **Time: 10:00 a.m to 01:00 p.m** **Maximum Marks: 80**

Instructions:
1. **Figures to right indicate marks.**
2. **Section wise separate answer book to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1	Attempt any **TWO** of the following	
A	Give the difference between Newtonian & non Newtonian flow. Classify various viscometers. Describe principle, working, advantages and disadvantages of cone & plate viscometer with suitable diagram.	**10**
B	Describe in brief types of non-Newtonian flow. How the thixotropy can be measured? Explain.	**10**
C	Write a note on Pharmaceutical Buffer.	**10**
Q 2	Attempt any **TWO** of the following	
A	Define sublimation, eutectic mixtures, polymorphism, amorphous and crystalline. Write a note on liquid Crystals.	**10**
B	Give the various methods for measurement of surface and interfacial tensions. Briefly explain principle, working, advantages and disadvantages of any one suitable method with suitable diagram.	**10**
C	Write a note on surface active agents. Discuss the electric properties of interfaces.	**10**

SECTION- II

Q 3	Attempt any **TWO** of the following	
A	Give the various methods for determination of particle size. Describe principle, working, advantages, disadvantages with labelled diagram of Optical microscopy.	**10**
B	Write a brief note on Micrometrics. Discuss the derived properties of powder.	**10**
C	Explain various type of Emulsion. Write a note on physical stability of emulsion.	**10**
Q 4	Attempt any **TWO** of the following	
A	Give definition, types, properties and application of suspension in Pharmacy. Differentiate flocculated suspension with deflocculated suspension.	**10**
B	Give definition, types, properties and application of colloids in Pharmacy. Briefly describe protective colloids.	**10**
C	I. Give identification test of Emulsion in Pharmacy II. Give theories of Emulsion formation.	**10**